전산화학 (2013년 1학기) 기말고사 정답

(총 **7**문제, **90**점)

1. **(총 10점, 각 5점)**

(a) $f(x,y,z) = 2x^2 + 3y^2 - z^2$ 이고,

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}$$

이므로

$$\begin{aligned}\nabla f(x,y,z) &= \mathbf{i}\frac{\partial f}{\partial x} + \mathbf{j}\frac{\partial f}{\partial y} + \mathbf{k}\frac{\partial f}{\partial z} \\ &= \mathbf{i}(4x) + \mathbf{j}(6y) + \mathbf{k}(-2z) \end{aligned} \tag{1}$$

(b) $f(x,y,z) = (x^2+y^2+z^2)^{-1/2}$ 이고,

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}$$

이므로

$$\begin{aligned}\nabla f(x,y,z) &= \mathbf{i}\frac{\partial f}{\partial x} + \mathbf{j}\frac{\partial f}{\partial y} + \mathbf{k}\frac{\partial f}{\partial z} \\ &= \mathbf{i}\left[-\frac{2x}{2(x^2+y^2+z^2)^{3/2}}\right] + \mathbf{j}\left[-\frac{2y}{2(x^2+y^2+z^2)^{3/2}}\right] + \mathbf{k}\left[-\frac{2z}{2(x^2+y^2+z^2)^{3/2}}\right] \\ &= -\frac{1}{(x^2+y^2+z^2)^{3/2}}(\mathbf{i}x + \mathbf{j}y + \mathbf{k}z) \end{aligned} \tag{2}$$

2. **(10점)** $\mathbf{A}$가 삼각행렬이므로, 행렬식의 값은 대각원소의 곱과 같다.

$$|\mathbf{A}| = \begin{vmatrix} 1 & 2 & 3 & -5 & -1 \\ 0 & 6 & 3 & 4 & -2 \\ 0 & 0 & 3 & -4 & -1 \\ 0 & 0 & 0 & -5 & -9 \\ 0 & 0 & 0 & 0 & -1 \end{vmatrix} = (1)(6)(3)(-5)(-1) = 90$$

3. **(10 점)**

$$\begin{aligned} 2x - 3y + 4z &= 8 \\ y - 3z &= -7 \\ x + 2y + 2z &= 11 \end{aligned}$$

위 식을 행렬을 이용해 나타내면

$$\begin{pmatrix} 2 & -3 & 4 \\ 0 & 1 & -3 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ -7 \\ 11 \end{pmatrix}$$

Cramer의 법칙을 이용해 위 연립방정식을 풀면,

$$x = \frac{\begin{vmatrix} 8 & -3 & 4 \\ -7 & 1 & -3 \\ 11 & 2 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & -3 & 4 \\ 0 & 1 & -3 \\ 1 & 2 & 2 \end{vmatrix}} = \frac{21}{21} = 1$$

$$y = \frac{\begin{vmatrix} 2 & 8 & 4 \\ 0 & -7 & -3 \\ 1 & 11 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & -3 & 4 \\ 0 & 1 & -3 \\ 1 & 2 & 2 \end{vmatrix}} = \frac{42}{21} = 2$$

$$z = \frac{\begin{vmatrix} 2 & -3 & 8 \\ 0 & 1 & -7 \\ 1 & 2 & 11 \end{vmatrix}}{\begin{vmatrix} 2 & -3 & 4 \\ 0 & 1 & -3 \\ 1 & 2 & 2 \end{vmatrix}} = \frac{63}{21} = 3$$

4. **(10점)**

$$\begin{aligned} \mathbf{a}\cdot\mathbf{b} &= \left(\frac{1}{2}\mathbf{q}_1 + \frac{1}{2}\mathbf{q}_2 + \frac{1}{2}\mathbf{q}_3 + \frac{1}{2}\mathbf{q}_4\right)\cdot\left(\frac{1}{2}\mathbf{q}_1 - \frac{1}{2}\mathbf{q}_2 + \frac{1}{2}\mathbf{q}_3 - \frac{1}{2}\mathbf{q}_4\right) \\ &= \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_1\cdot\mathbf{q}_4 + \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_2\cdot\mathbf{q}_4 \\ &\quad + \frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_3\cdot\mathbf{q}_4 + \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_1 - \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_2 + \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_3 - \frac{1}{4}\mathbf{q}_4\cdot\mathbf{q}_4 \\ &= \frac{1}{4} - \frac{1}{4} + \frac{1}{4} - \frac{1}{4} = 0, \quad (\mathbf{q}_i\cdot\mathbf{q}_j = \delta_{ij} \text{ 이므로}) \end{aligned}$$

그러므로 **a**와 **b**는 서로 직교한다.

5. **(총 10점, 각 5점)**

(a)

$$\boldsymbol{\sigma}_1^2+\boldsymbol{\sigma}_2^2+\boldsymbol{\sigma}_3^2 = \begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}+\begin{pmatrix}0&-i\\i&0\end{pmatrix}\begin{pmatrix}0&-i\\i&0\end{pmatrix}+\begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}$$
$$= \begin{pmatrix}1&0\\0&1\end{pmatrix}+\begin{pmatrix}1&0\\0&1\end{pmatrix}+\begin{pmatrix}1&0\\0&1\end{pmatrix} = \begin{pmatrix}3&0\\0&3\end{pmatrix}$$

(b)

$$\boldsymbol{\sigma}_1\boldsymbol{\sigma}_2 = \begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0&-i\\i&0\end{pmatrix} = \begin{pmatrix}i&0\\0&-i\end{pmatrix} = i\boldsymbol{\sigma}_3 \tag{3}$$

6. **(총 10점, 각 5점)**

(a)

$$\begin{aligned}\left[\frac{d^2}{dx^2},x\right]f &= \frac{d^2}{dx^2}(xf)-x\frac{d^2}{dx^2}f\\ &= \frac{d}{dx}(f+xf')-xf''\\ &= f'+f'+xf''-xf'' = 2f' = 2\frac{df}{dx}\end{aligned}$$

$$\left[\frac{d^2}{dx^2},x\right] = 2\frac{d}{dx}$$

(b)

$$\begin{aligned}\left[\frac{d}{dx}-x,\frac{d}{dx}+x\right]f &= \left(\frac{d}{dx}-x\right)\left(\frac{d}{dx}+x\right)f-\left(\frac{d}{dx}+x\right)\left(\frac{d}{dx}-x\right)f\\ &= \frac{d^2f}{dx^2}+\frac{d}{dx}(xf)-x\frac{df}{dx}-x^2f-\frac{d^2f}{dx^2}+\frac{d}{dx}(xf)-x\frac{df}{dx}+x^2f\\ &= f''+f+xf'-xf'-x^2f-f''+f+xf'-xf'+x^2f = 2f\end{aligned}$$

$$\left[\frac{d}{dx}-x,\frac{d}{dx}+x\right] = 2$$

7. **(총 30점)**

(a) **(5점)**

$$\begin{aligned}\big[\hat{x},\hat{p}\big]f = (\hat{x}\hat{p}-\hat{p}\hat{x})f &= x\frac{1}{i}\frac{d}{dx}f-\frac{1}{i}\frac{d}{dx}(xf)\\ &= \frac{1}{i}\big(xf'-f-xf'\big)\\ &= -\frac{1}{i}f = if\end{aligned} \tag{4}$$

그러므로

$$\boxed{\big[\hat{x},\hat{p}\big] = i} \tag{5}$$

(b) **(10점)**

$$\begin{aligned}
[\hat{a}_-, \hat{a}_+] &= \hat{a}_-\hat{a}_+ - \hat{a}_+\hat{a}_- \\
&= \left\{\frac{1}{\sqrt{2}}(\hat{x}+i\hat{p})\right\}\left\{\frac{1}{\sqrt{2}}(\hat{x}-i\hat{p})\right\} - \left\{\frac{1}{\sqrt{2}}(\hat{x}-i\hat{p})\right\}\left\{\frac{1}{\sqrt{2}}(\hat{x}+i\hat{p})\right\} \\
&= \frac{1}{2}\big(\hat{x}^2 - i\hat{x}\hat{p} + i\hat{p}\hat{x} + \hat{p}^2\big) - \frac{1}{2}\big(\hat{x}^2 + i\hat{x}\hat{p} - i\hat{p}\hat{x} + \hat{p}^2\big) \\
&= -i(\hat{x}\hat{p} - \hat{p}\hat{x}) = -i\big[\hat{x}, \hat{p}\big] = 1 \quad \text{((5) 식의 결과로부터)}
\end{aligned}$$

(c) **(15점)**

$$\begin{aligned}
\hat{a}_+\hat{a}_- &= \left\{\frac{1}{\sqrt{2}}(\hat{x}-i\hat{p})\right\}\left\{\frac{1}{\sqrt{2}}(\hat{x}+i\hat{p})\right\} \\
&= \frac{1}{2}\big(\hat{x}^2 + i\hat{x}\hat{p} - i\hat{p}\hat{x} + \hat{p}^2\big) \\
&= \frac{1}{2}\big(\hat{x}^2 + i[\hat{x}, \hat{p}] + \hat{p}^2\big) \\
&\quad \text{(5) 식으로부터} \\
&= \frac{1}{2}\big(\hat{x}^2 + \hat{p}^2 - 1\big) \qquad (6)
\end{aligned}$$

(6) 식의 결과로부터

$$\frac{1}{2}\hat{x}^2 + \frac{1}{2}\hat{p}^2 = \hat{a}_+\hat{a}_- + \frac{1}{2} \qquad (7)$$

(7) 식의 결과로부터 $\hat{H}$ 는

$$\boxed{\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\hat{x}^2 = \hat{a}_+\hat{a}_- + \frac{1}{2}}$$